

- A. What is the cost variance, schedule variance, cost performance index (CPI), and schedule performance index (SPI) for the project?

CV, SV, CPI, SPI

PV = \$100,000

EV = \$90,000

AC = \$115,000

BAC = \$200,000

Cost Variance (CV) = EV - AC = \$90,000 - \$115,000 = -\$25,000

Schedule Variance (SV) = EV - PV = \$90,000 - \$100,000 = -\$10,000

Cost Performance Index (CPI) = EV/AC = 90,000 / 115,000 = 0.783

Schedule Performance Index (SPI) = EV/PV = 90,000 / 100,000 = 0.90

- B. How is the project doing? Is it ahead of schedule or behind schedule? Is it under budget or over budget?

Schedule: SPI < 1 → behind schedule (earned less value than planned).

Cost: CPI < 1 → over budget (spending more than value earned).

- C. Use the CPI to calculate the estimate at completion (EAC) for this project. Is the project performing better or worse than planned?

Estimate at Completion (EAC) – BAC/CPI = 200,000/0.783 = 255,427

Worse than planned (forecast exceeds the \$200k budget by -\$55.4k)

- D. Use the SPI to estimate how long it will take to finish this project.

To forecast the total duration using the Schedule Performance Index (SPI):

Revises duration = Original Duration/SPI

= 6/0.90 = 6.67 months (so the project will finish 20 days later than planned (0.67x30days=20days))

- E. Sketch an earned value chart using the preceding information. See Figure 7-5 as a guide

